VR Graffiti



Why Learn Graffiti in VR?

Graffiti is a physical skill, not just a visual art.

Learning graffiti requires coordination between:

Movement

controlling hand trajectories and smooth strokes.

Distance & Speed

adjusting spray width, opacity, and consistency.

Lettering

understanding letter structure, proportions, and styles.

The problem

Beginners face barriers to practicing consistently, including:



Limited access to appropriate spaces



Material costs



Safety considerations

The opportunity

VR can provide an accessible, repeatable environment for learning and experimenting with graffiti techniques.

Wall: Singlespeedfahrer. Cans: Chris Barbalis (CC0). Mask: Haragayato (CC BY-SA 3.0). Monochrome display; cans cropped.

Graffiti combines artistic expression with embodied motor skills, but beginners rarely have access to structured, repeatable practice with feedback.

Existing Solutions & The Gap



Kingspray Graffiti VR

Realistic spray physics, multiple caps, multiplayer painting, and reference projection.

Tutorials focus on application features rather than graffiti techniques or artistic fundamentals.

Gap: Primarily a creative simulation tool, better suited to experienced artists than complete beginners.



Vermillion VR

Realistic oil painting, guided tutorials, reference tracing, and color mixing.

Gap: Focuses on traditional painting rather than graffiti-specific techniques and lettering.

Our Opportunity

A beginner-oriented VR learning environment with structured instruction, progressive skill development, and measurable feedback.

Kingspray, official PC description. Vermillion, official website. Images supplied by presenter. Gap statements reflect our project analysis.

From Simulation to Skill Learning

Observe → Trace → Reproduce → Remix

Our learning experience draws on three educational theories:

Observational Learning

Observe and imitate an expert's hand movements through an animated virtual hand.

Scaffolding

Gradually remove visual guides and assistance as learners gain proficiency.

Constructionism

Develop understanding by creating personalized graffiti and experimenting with different styles.

From imitation to independent creation.

A first lesson: Bubble S

From guidance to independence.



Trace with spatial cues that gradually fade

Observe

See pose and pace.

Trace

Follow the motion trail.

Reproduce

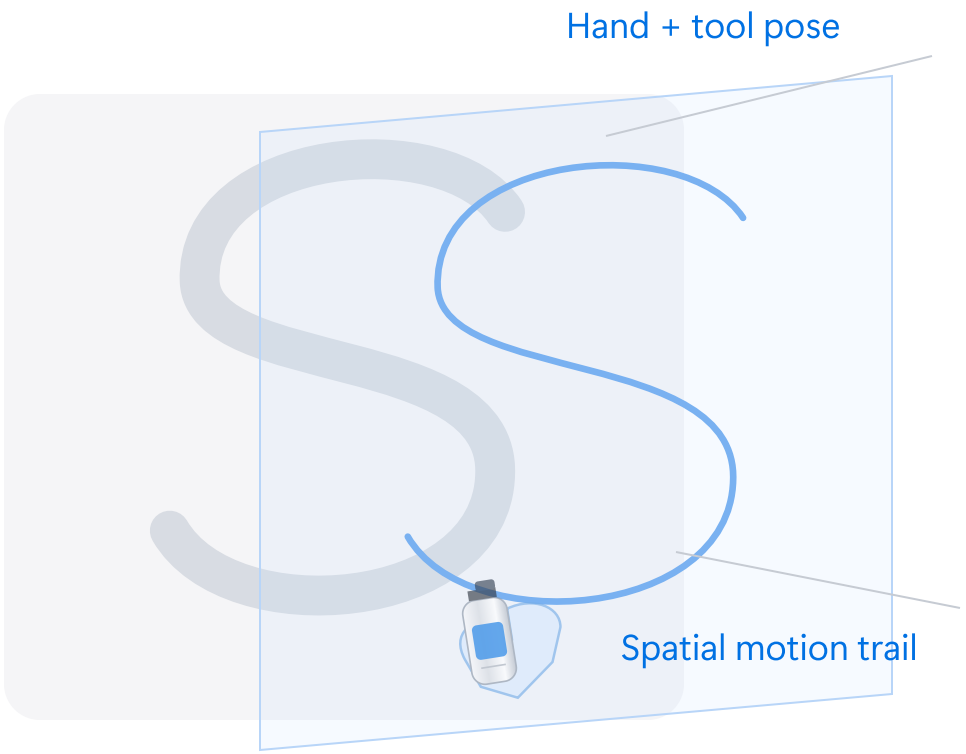
Paint without the overlay.

Remix

Make it your own.

Concept visualization. Support can be requested again. Remix has no single correct trajectory.

Guidance that steps back



Observational learning

See stroke order
and timing.

Scaffolding and fading

Full guide → faint trail
→ independent attempt.

Feedback and reflection

Distance cues during practice.
Path comparison afterward.

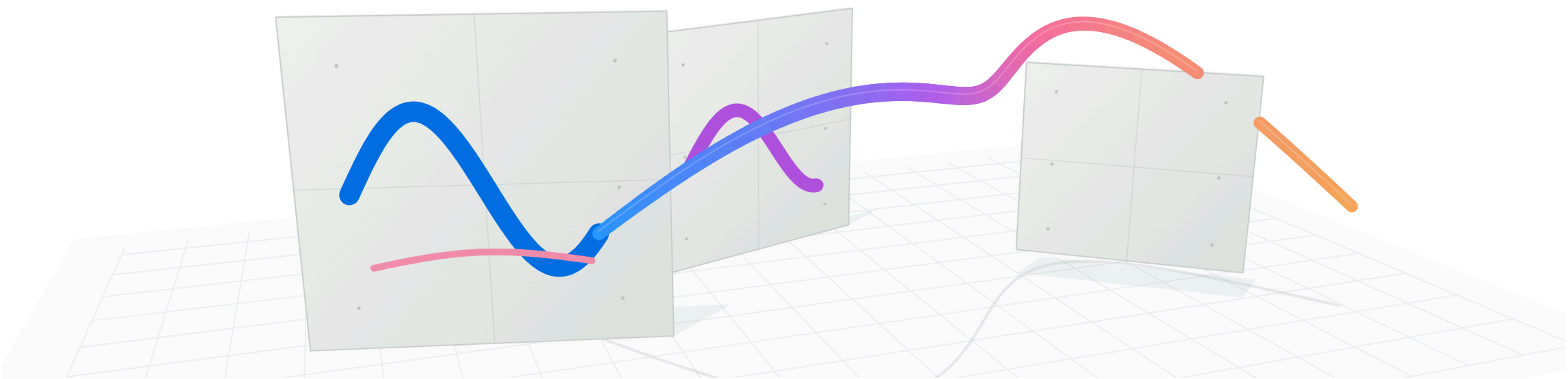
Embodied interaction

Move, paint and follow cues
in the same space.

Han et al., 2022. Wulf et al., 2010. Yu et al., CHI 2024. Levac et al., 2019. Design hypotheses, not established outcomes for this application.

Free Space to Practice

Shape your space. Create on your own terms.



Paint across surfaces and into open space

Create surfaces

Add walls of different sizes.

Arrange your space

Move and rotate your layout.

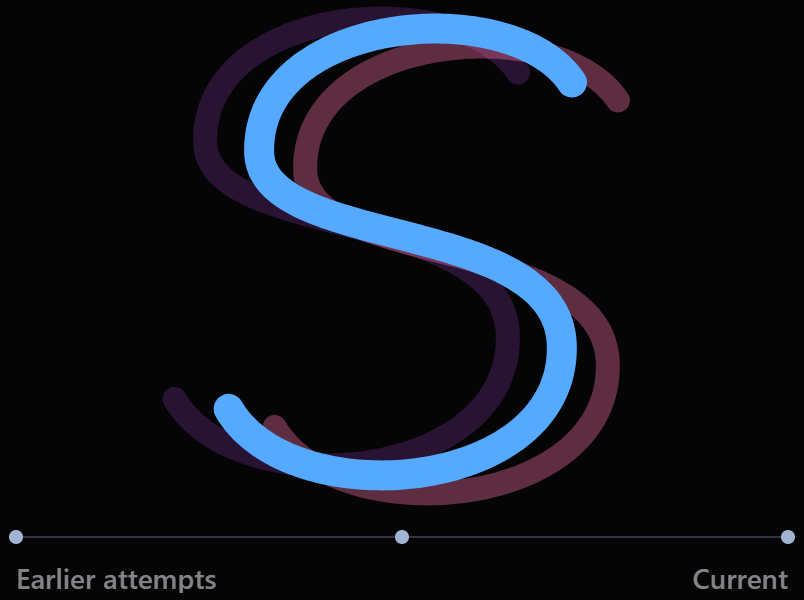
Go beyond walls

Extend your artwork into open space.

Concept animation: create surfaces, arrange the space, then paint across and beyond them.

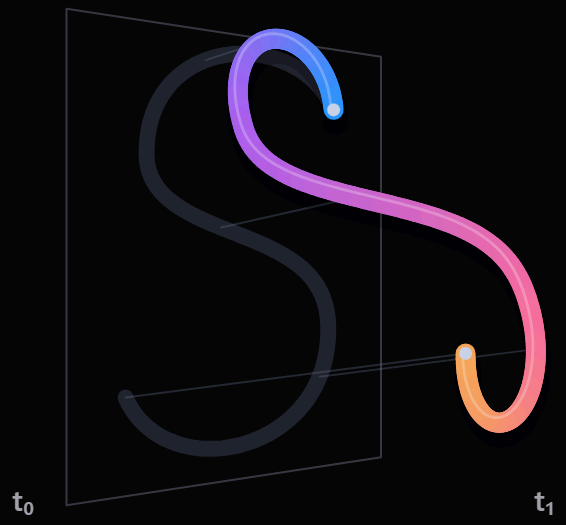
Beyond Traditional 3D Art

Create with time, motion and recorded gestures.



Temporal Layers

Past attempts.
One composition.



Time mapped to spatial depth

Time-to-Space

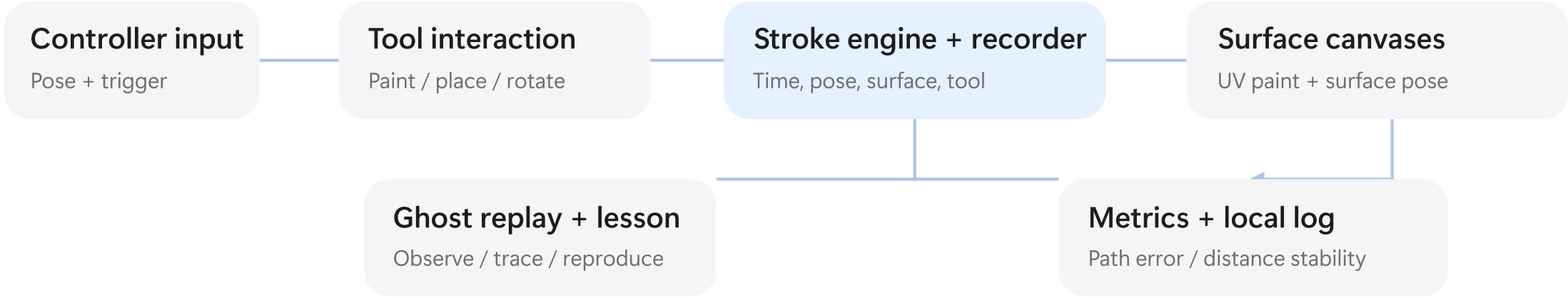
Time becomes depth.
The stroke leaves the wall.

Beyond the MVP. Both reuse recorded gestures.

Concept studies. Existing replay and spatial painting inform this temporal extension.

One recording. Several uses.

Meta Quest + controllers · Unity + C# · OpenXR + XR Interaction Toolkit



Paint + GPU cost

Profile brush stamps on Quest.
Cap history and lower resolution if needed.

Replay alignment

Use wall coordinates.
Calibrate height and reach.
Start with one plane.

Fidelity + scope

Visual distance feedback.
Protect one spray lesson.
Keep contact tools basic.

Unity, XR input options. Proposed local architecture. No backend, fluid simulation or fast hand-tracking dependency.

One lesson.
One semester.

Smart Chen · Solo developer

Experience: Unity XR, interaction, visual design and media.

Owns design, code, art direction, logging and evaluation.

Phase 1

Foundations

Quest input + studio.
Create + arrange walls.

Phase 2

Tools + lesson

Basic marker + roller.
Bubble S + lesson UI.

Phase 3

Guided practice

Record + replay.
Ghost trail + fading cues.

Phase 4

Test + polish

Unguided attempt + logs.
Usability + stable demo.

Semester MVP

Editable walls. Three tools. One S lesson.
Guided practice and free creation.

Stretch goals

Spatial painting. Temporal Layers.
Time-to-Space. Optional pilot.

Relative roadmap; dates follow the course deadline. Concept stage: no prototype or participant results.

References

Research informs the design; outcomes remain untested.

Han, Y., Ali, S. K. B. S., & Ji, L. (2022)

Use of Observational Learning to Promote Motor Skill Learning in Physical Education: A Systematic Review.
IJERPH, 19(16), 10109. 18 included studies.

Levac, D. E., Huber, M. E., & Sternad, D. (2019)

Learning and transfer of complex motor skills in virtual reality: a perspective review.
Journal of NeuroEngineering and Rehabilitation, 16, 121.

Yu, X., Lee, B., & Sedlmair, M. (2024)

Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems.
CHI 2024.

Wulf, G., Shea, C., & Lewthwaite, R. (2010)

Motor skill learning and performance: a review of influential factors.
Medical Education, 44(1), 75–84.

Infectious Ape. Kingspray Graffiti VR

Official PC listing: spray, replay, projection and tutorials.
Features vary by platform.

Open Brush

Official project site. Room-scale 3D painting.

Krylon. How to Spray Paint

Manufacturer guidance: technique and ventilation.

Unity. XR input options

Official Unity 6 documentation.
OpenXR, controller tracking and XR Interaction Toolkit.

Product photography: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography.

Original sources linked. Product documentation checked October 27, 2026. Q&A backup.

Discussion

Questions for the next iteration

Why VR instead of a video?

Show tool motion at the wall.
Practice in the same space.

Will this transfer to real paint?

Unknown: controllers lack material feedback.
Physical transfer needs a separate test.

Why keep three tools?

Shared paint and recording systems.
Spray first; basic contact tools.

What if the plan is too large?

Protect the spray lesson and logging.
Defer the pilot and temporal art.

How will remix be assessed?

Reflection or artist feedback.
Path error measures copying.

What is the first thing to validate?

Can beginners follow the cues
while seeing their own mark?

Q&A backup. Physical transfer and comparative learning benefits remain untested.

Three tools. Different control.

Spray, draw and fill.



Montana BLACK 400 ml

Spray can



Distance shapes the footprint



MTN Street Paint 15 mm

Paint marker



Contact gives a crisp edge



Purdy Mini Roller System

Paint roller



Broad passes build coverage

Controllers approximate motion; pressure, friction and weight are absent.

Product photos: Montana Cans product photography. MTN / Spray Planet product photography. Purdy product photography. Motion illustrations show proposed virtual behavior.